

Geometry supervision for 3DGRUT

Depth and normal quality on OB3D: measurements

7k iterations, OB3D, three scenes

Eight experiments across five candidate supervision terms, each reported as

PREMISE EXPERIMENT RESULT QUALITATIVE CONCLUSION

Setup

SCENES OB3D, with reference depth and normals:

sponza, lone-monk, emerald-square.

PROTOCOL 7k iterations. Reference geometry is read by evaluation only; no training term uses it. Held-out split for all metrics.

SEED NOISE over `seed_initialization` 1–4:

- PSNR ± 0.04 – 0.13 dB
- abs_rel ± 0.002 – 0.004
- normal ± 0.1 – 1.2°

Single-term cells are 3-seed; combination cells are single-seed.

METRICS

- abs_rel — mean $|d - d^*|/d^*$ over pixels where the reference has depth and the ray is opaque enough to have one. **Offline diagnostics use a stricter mask than this**, so their numbers are not comparable to these — which cost two experiments (Experiment 8).
- δ_1 — fraction of those pixels within 25% of the reference; reported as the *failure* rate $1 - \delta_1$.
- normal° — mean angle to the reference normal, signed (a flipped normal scores 180° , not 0°).
- n_gain — **control**: normal° of the view-direction control minus normal° of the render. The control points every normal back along the view ray and uses no geometry.
- PSNR — on the composited image, as validation reports it.

$\text{n_gain} < 0$: the normal buffer is worse than using no geometry at all.

Reference runs: both primitives, no geometry term

Run	sponza				lone-monk				emerald-square			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92

RESULT

- PSNR is within 0.03 dB between primitives on every scene.
- gaussian has the better abs_rel on sponza and lone-monk; trisurfel on emerald-square.
- trisurfel has the better normals on **all three**, by 7.7–19.3^o.
- gaussian's n_gain is -9.6 to -19.1. trisurfel's is -1.9 to +0.9.

CONCLUSION

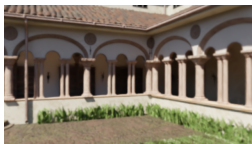
- Neither primitive's normal buffer beats the view-direction control by a usable margin, at equal PSNR.
- The **best reference** is per scene *and* per metric, so no single row can serve as the baseline. Every following table carries both.

Reference runs, qualitative: RGB and depth

QUALITATIVE lone-monk, held-out frame. Depth in turbo over a shared range, near to far.



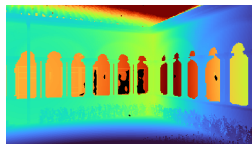
reference RGB



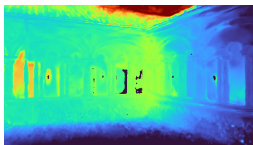
gaussian 36.72 dB



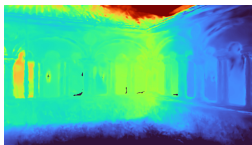
trisurfel 36.71 dB



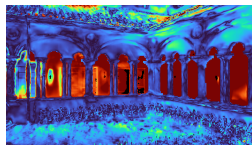
reference depth



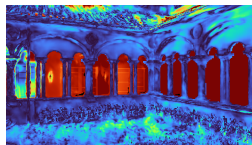
gaussian depth



trisurfel depth



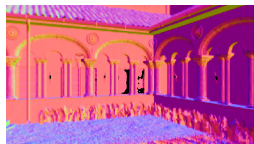
gaussian rel. error



trisurfel rel. error

Reference depth reads the arch openings as **far** (orange). Both references put a surface across them at wall depth (green), and both error maps saturate there. Error maps are capped at the `delta1` boundary, so a saturated pixel is one the metric counts as wrong.

Reference runs, qualitative: normals



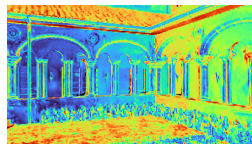
reference normals



gaussian 54.3° , $n_gain -19.1$



trisurfel 35.0° , $n_gain +0.2$



trisurfel angular error

RESULT

- **gaussian**: high-frequency noise, no recoverable surface structure.
- **trisurfel**: smooth and coherent, **but nearly uniform in hue** — most normals point in one direction. That direction is close to the view ray, which is what $n_gain = +0.2$ reports: 19° better than **gaussian** in raw angle, and still no better than the geometry-free control.

CONCLUSION The raw angle and the visual impression both rank **trisurfel** far ahead; the control says neither reference has a normal buffer carrying geometry. Raw angle alone would have declared this solved.

The three scenes fail differently

Scene	floaters px	δ_1 failures	Failure mode
spnza	0.2–2%	moderate	floaters; per-ray spread detects them (AUC 0.68–0.90)
emerald-square	up to 13%	high	floaters, largest depth range
lone-monk	0.01%	17.7%	confidently placed, wrong; no floaters to find

RESULT `lone-monk` has the fewest floaters and the worst δ_1 rate. Its rays are already concentrated, at the wrong distance — visible on the previous slide as an opaque surface across the arch openings.

CONCLUSION A term that works by concentrating a ray cannot help `lone-monk`. Metrics are reported per scene throughout; a scene mean would average this away.

Experiment 1: depth-normal consistency

PREMISE The rendered normal and the rendered depth are produced independently and nothing ties them together. The depth gradient implies a surface orientation; the normal buffer need not agree with it.

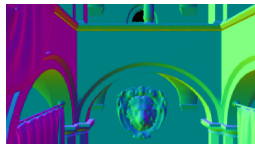
EXPERIMENT Penalise the angle between the rendered normal and the normal implied by the local geometry of the rendered depth. No external data; internal consistency only. Weight 0.05, `depth_normal_from_iter` lowered from 7000 to fit a 7k sweep.

RESULT

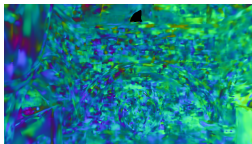
Run	<i>sponza</i>				<i>lone-monk</i>				<i>emerald-square</i>			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92
dn 0.05	0.0474	25.4	+17.4	36.10	0.0976	37.4	-2.1	36.56	0.1309	41.3	+9.0	32.42

Experiment 1: qualitative and conclusion

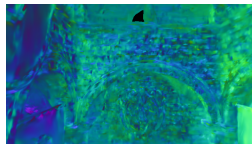
QUALITATIVE `sponza` normals, against the reference and both reference runs.



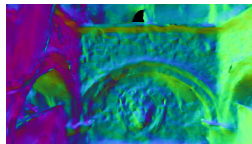
reference



`gaussian` 52.4°



`trisurfel` 44.7°



`dn` 0.05 25.4°

CONCLUSION

- Normals: beats `gaussian` on all three scenes by 17–27°. Against the *best* reference it wins on `sponza` (−19.3°) and `emerald-square` (−8.1°) and **loses on** `lone-monk` (37.4° against `trisurfel`'s 35.0°).
- `n_gain` turns positive on `sponza` and `emerald-square`; on `lone-monk` it stays **negative** (−2.1), as `gaussian`'s does.
- Depth: −17.0% on `sponza`, **under 1%** on `lone-monk` — i.e. nothing on the scene whose depth is worst.
- Cost: 0.16–0.50 dB PSNR.

Landed, default off. The arch, masonry courses and boss are recovered from a buffer that was noise. But `lone-monk` resists it on both metrics, and had the deck compared only against `gaussian` this would have read as a clean win on all three scenes.

Experiment 2: depth variance along the ray

PREMISE A ray that spreads its weight over many distances has no single surface. Penalising the per-ray variance of depth should make each ray commit to one.

EXPERIMENT Render per-ray M_2 , form $\text{Var} = M_2 - D^2/\text{acc}$, penalise it. Then the scale-free form $\text{Var}/\mu^2 = \text{acc} \cdot M_2/D^2 - 1$, which needed no new buffer, kernel or backward pass. Weights 1 and 0.01.

RESULT

Run	sponza				lone-monk				emerald-square			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92
dvrel 1	0.0593	51.8	-8.9	35.27	0.0980	56.1	-20.9	36.65	0.1792	67.6	-17.3	31.13
dvrel 0.01	0.0526	52.4	-9.7	36.33	0.0993	53.2	-18.0	36.80	0.1325	65.7	-15.4	32.87

The absolute form (earlier sweep, not in this table) gave +6% abs_rel on sponza and raised the floater fraction from 0.30% to 1.62%.

From depth_variance_mechanism.py: per-ray spread $\times 0.17$ while wrong | tight rose $\times 170$.

Experiment 2: mechanism and conclusion

WHY In pairwise form,

$$M_2 - \frac{D^2}{\text{acc}} = \frac{1}{2 \text{acc}} \sum_{ij} w_i w_j (t_i - t_j)^2$$

- Variance is zero for a Dirac at *any* distance: the objective says commit, never where.
- Two-hit case is a **double well** in the near opacity, since $w_{\text{near}} w_{\text{far}}$ carries $\alpha(1 - \alpha)$.
- The near well is **absorbing**: at $\alpha = 1$ transmittance zeroes the gradient to everything behind, for every loss. The error is permanent.

CONCLUSION

- **Refuted as posed**: the absolute form did what it was asked — rays six times tighter — and became 170 times more likely to be confidently wrong.
- The relative form at 0.01 improves depth on *sponza* (**−7.9%**) at no PSNR cost, but against the best reference it is **null** on *emerald-square* (−0.2%) and **worse** on *lone-monk* (+1.2%). Normals untouched.
- Identical minima, different behaviour: descent responds to the barrier, not the location of the optima.

RESCALING Dividing by μ^2 leaves both wells at exactly zero, and was first dismissed for that reason.

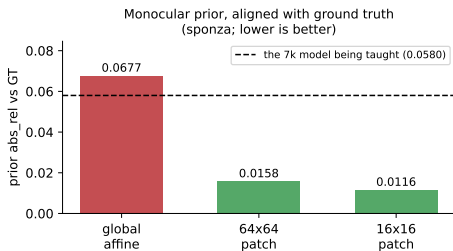
	absolute	relative
separatrix	0.49	0.82
bad basin	51%	18%
sponza	+6%	−7.9%

Experiment 3: regression against a monocular prior

PREMISE A monocular depth model carries information the scene does not. Plan: supervise it with a scale-invariant loss, fitting per-image scale and shift so the predictor's arbitrary affine gauge is not supervised.

EXPERIMENT Before training anything: measure the prior itself. DepthAnythingV2-Base on sponza, affine-aligned against *ground truth* at decreasing granularity — the most favourable alignment possible, since it uses the reference the loss would not have.

RESULT



- One affine per frame: **0.0677**, against **0.0580** for the 7k model it would be teaching.
- Per 64×64 patch: 0.0158. Per 16×16 : **0.0116**.
- Correlation with $1/z$ is +0.978; an affine fit reaches R^2 0.956 in disparity against 0.821 in depth.
- The prior emits z-depth; this renderer's convention is Euclidean ray distance. Radial divergence reaches 20% at the image corners.

CONCLUSION **Abandoned before training** — and **wrongly**. Experiment 8 overturns this: both the scene and the pixel mask behind “0.0677 against 0.0580” were the problem.

Experiment 4: ordinal supervision from the same prior

PREMISE Only the prior's *ordering* survived the diagnostic. On pixels where the trained model fails `delta1`, the prior is closer to the reference 91% of the time.

EXPERIMENT Read the ordering only. For a pixel pair, penalise the rendered gap only where render and prior disagree about which is nearer:

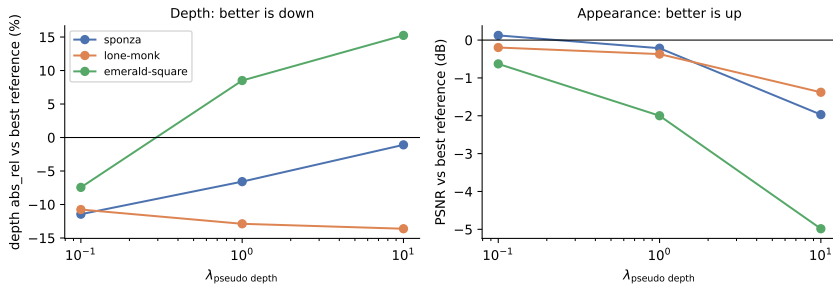
$$\mathcal{L} = \frac{1}{|\mathcal{P}|} \sum_{(a,b) \in \mathcal{P}} \text{relu}(- (d_a - d_b) \cdot \text{sign}(\text{disp}_b - \text{disp}_a)) / \text{scene extent}$$

- Invariant to *any* increasing transform of the prior: disposes of alignment, the disparity inversion and z-vs-ray-distance at once.
- One-sided: a correctly ordered pair contributes zero and no gradient.
- Pairs formed by *cropping* at a random offset $\leq 5\%$ of the image side, not `torch.roll`, which would wrap opposite edges into pairs.

RESULT

Run	sponza				lone-monk				emerald-square			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92
pd 0.1	0.0506	52.5	-9.6	36.41	0.0876	53.1	-17.8	36.52	0.1228	66.2	-15.9	32.29
pd 1	0.0534	52.8	-9.9	36.07	0.0855	53.3	-18.1	36.35	0.1440	69.3	-19.0	30.92
pd 10	0.0565	55.0	-12.2	34.32	0.0848	57.0	-21.8	35.34	0.1529	69.3	-19.0	27.94

Experiment 4: weight sweep



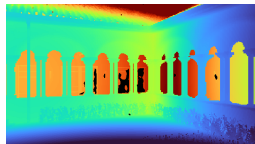
RESULT

- $\lambda = 0.1$: depth -7.4 to -11.5% against the best reference on **all three** scenes, at $>7\sigma$ on each (3-seed means, pooled standard error). First term in this report to improve depth everywhere.
- $\lambda = 1$: more depth on `lone-monk` (-12.9%), but `emerald-square` depth goes **positive** and PSNR costs **2.0 dB**.
- $\lambda = 10$: most of the depth gain given up; $\lambda = 100$ diverges.
- Normals unchanged at every weight — expected, since the rendered normal is the particle's local z axis and depth ordering does not reach it.

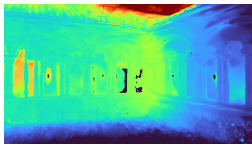
CONCLUSION Usable band is narrow and centred on $\lambda = 0.1$. Cost there: **+0.12 dB** on `sponza`, **-0.20** and **-0.63 dB** on the others.

Experiment 4: qualitative

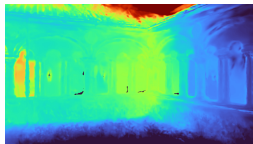
QUALITATIVE lone-monk depth and relative error, against the reference and both reference runs.



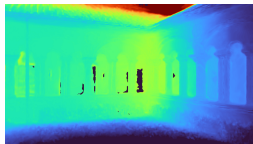
reference depth



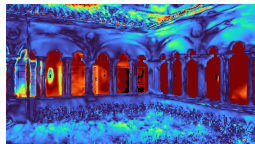
gaussian



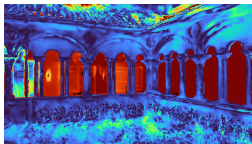
trisurfel



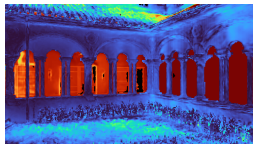
pd 0.1



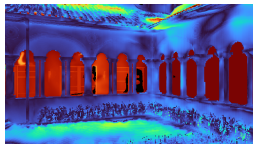
gaussian error



trisurfel error



pd 0.1 error



pd 0.1 + dn error

CONCLUSION

- The arch openings stay **saturated** in every render: the ordinal term does **not** fix them.
- What improves is the diffuse error — roof line, walls, ground all go blue.
- Consistent with the mechanism: pairs are drawn at $\leq 5\%$ offsets and the prior is reliable locally (0.0116 per patch), not globally (0.0677). A hole in a wall is a long-range discontinuity.

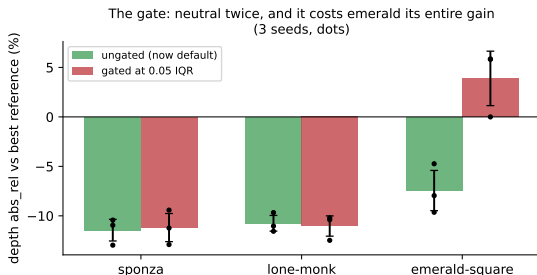
Experiment 5: gating pairs by the prior's confidence

PREMISE Near-tied pairs are where the prior's sign is noise. Offline, on *sponza*:

- ordinal agreement with the reference, all pairs: 84%
- after dropping pairs whose disparity gap is below 0.05 of the map's IQR: **97%**

So ungated, roughly one pair in six pushes the wrong way.

EXPERIMENT Same term, same weight, gate on and off. 3 seeds each.



RESULT

Run	sponza				lone-monk				emerald-square			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92
pd 0.1	0.0506	52.5	-9.6	36.41	0.0876	53.1	-17.8	36.52	0.1228	66.2	-15.9	32.29
pd 0.1 gated	0.0508	52.6	-9.8	36.35	0.0873	54.3	-19.1	36.73	0.1378	67.6	-17.3	32.28

Experiment 5: conclusion

RESULT Neutral on `sponza` and `lone-monk` (within seed noise); on `emerald-square` it removes the entire gain, -7.4% becoming $+3.9\%$.

CONCLUSION **Refuted**. Two reasons, both derivable from numbers already collected before the runs:

- A large disparity gap selects a **long-range** comparison — precisely where this prior is weakest (0.0677 global against 0.0116 per patch). The gate keeps the prior's worst structure and discards its best.
- For a **one-sided** loss, 97% agreement means only 3% of surviving pairs can produce any gradient at all. High agreement is a warning, not a benefit.

The offline metric counted pairs instead of asking what they teach. Default is now off, pinned by a test so re-enabling it must be re-measured.

Experiment 6: combining terms

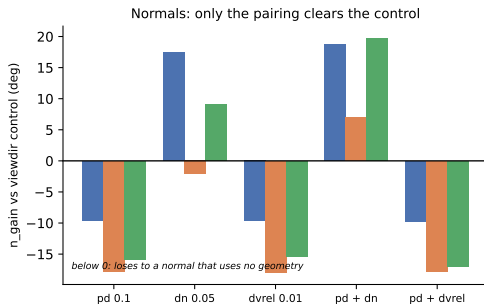
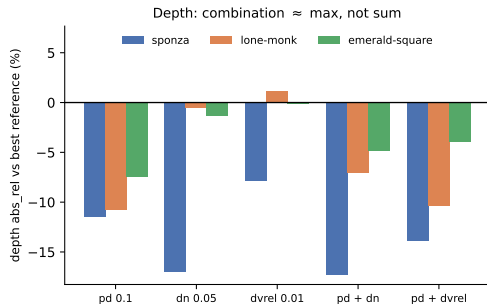
PREMISE The terms target different failures: internal consistency ($_{dn, dvrel}$) against an external prior ($_{pd}$). Per the failure-mode table they should be complementary by scene.

EXPERIMENT Each term at its own measured optimum, and the pairs. Single seed.

RESULT

Run	<i>sponza</i>				<i>lone-monk</i>				<i>emerald-square</i>			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
gaussian (ref)	0.0571	52.4	-9.6	36.29	0.0982	54.3	-19.1	36.72	0.1368	65.8	-15.5	32.90
trisurfel (ref)	0.0600	44.7	-1.9	36.19	0.0987	35.0	+0.2	36.71	0.1327	49.4	+0.9	32.92
pd 0.1	0.0506	52.5	-9.6	36.41	0.0876	53.1	-17.8	36.52	0.1228	66.2	-15.9	32.29
dn 0.05	0.0474	25.4	+17.4	36.10	0.0976	37.4	-2.1	36.56	0.1309	41.3	+9.0	32.42
pd 0.1 + dn 0.05	0.0472	24.0	+18.8	35.77	0.0912	28.3	+6.9	36.50	0.1262	30.5	+19.8	31.97
dvrel 0.01	0.0526	52.4	-9.7	36.33	0.0993	53.2	-18.0	36.80	0.1325	65.7	-15.4	32.87
pd 0.1 + dvrel 0.01	0.0492	52.6	-9.8	36.16	0.0880	53.1	-17.8	36.80	0.1275	67.3	-17.0	32.15

Experiment 6: result

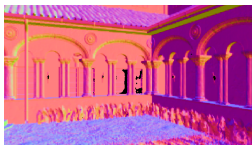


RESULT

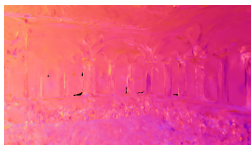
- **Depth is sub-additive.** Pairs land near the *max* of their parts, never the sum. On *lone-monk*, *pd + dn* (-7.1%) is **worse than *pd* alone** (-10.8%).
- **Normals are super-additive.** *pd* alone does nothing for normals; *dn* alone leaves *lone-monk* below the control. *pd + dn* is the **only** configuration whose normals clear the control on all three scenes, and it beats *trisurfel* by $6.7\text{--}20.7^\circ$.
- Mechanism: *dn* ties the normal to the depth *gradient*, so it can only be as good as the depth given to it. The prior improves the depth; consistency carries that into normals.

Experiment 6: qualitative

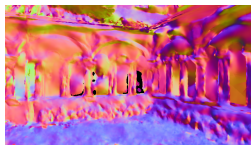
QUALITATIVE lone-monk normals. Same frame, same encoding.



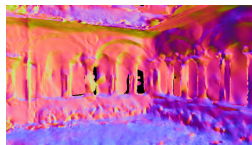
reference



trisurfel 35.0° , $+0.2$



dn 0.05 37.4° , -2.1



pd + dn 28.3° , $+6.9$

CONCLUSION

- dn alone: columns and arches blobby; still **below the control**.
- pd + dn: colonnade resolved, and the only panel here whose n_{gain} is meaningfully positive. Adding a depth term that touches no normals is what flipped the sign.

Experiment 6: the weight trap

EXPERIMENT This pairing was first measured with $\lambda = 1$ for **both** terms — $10\times$ and $100\times$ their measured optima — and written up as a null. **RESULT** `abs_rel`, and the change against each scene's best reference run.

Run	sponza		lone-monk		emerald-square	
gaussian (ref)	0.0571		0.0982		0.1368	
trisurfel (ref)	0.0600		0.0987		0.1327	
pd 0.1	0.0506	−11.4%	0.0876	−10.8%	0.1228	−7.4%
dvrel 0.01	0.0526	−7.9%	0.0993	+1.2%	0.1325	−0.2%
pd 0.1 + dvrel 0.01	0.0492	−13.9%	0.0880	−10.4%	0.1275	−3.9%
pd 1 + dvrel 1	0.0528	−7.6%	0.0855	−12.9%	0.1510	+13.8%

At their own optima the verdict changes from *harmful* to *scene-dependent*: the pair beats both single terms on `sponza`, is a wash on `lone-monk`, and is still worse than `pd` alone on `emerald-square`.

CONCLUSION A null from a combination measured at the wrong weights is not a null about the combination. The over-weighted cell is kept in the sweep so the trap stays visible.

Experiment 7: is the *prior* the limiting factor?

PREMISE Experiments 3–6 varied the loss and held the prior fixed at `DepthAnythingV2-Base`. The ordinal term reads only the prior's ordering, so a prior that orders pixels better should train better — and Depth Anything 3 rebuilt the model, not just the weights: it predicts *depth* where V2 predicts disparity.

EXPERIMENT Swap the prior, change nothing else. Same $\lambda = 0.1$ ordinal term, same `trisurfel` primitive, 3 seeds. The backend now records which *quantity* it emits, and the loss takes its sign from that, so the two priors cannot be compared with the sign inverted — a mistake that is invisible in the loss curve.

RESULT

Run	<i>sponza</i>				<i>lone-monk</i>				<i>emerald-square</i>			
	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR	abs_rel	normal ^o	n_gain	PSNR
trisurfel (ref)	0.0618	44.7	−1.9	36.10	0.0998	35.2	+0.0	36.78	0.1319	49.2	+1.1	32.81
pd 0.1, DAV2-Base	0.0538	44.4	−1.6	36.14	0.0865	34.6	+0.6	36.68	0.1250	48.1	+2.2	31.85
pd 0.1, DA3MONO-L	0.0522	44.5	−1.7	36.15	0.0866	34.5	+0.7	36.61	0.1108	48.0	+2.3	32.20

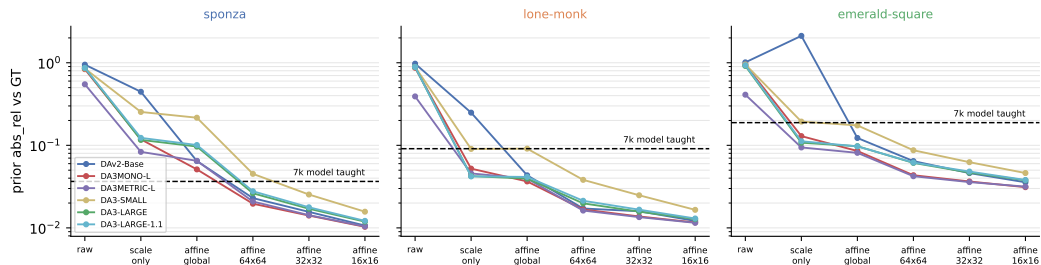
- *emerald-square*: depth gain $-5.3\% \rightarrow -16.0\%$, a factor of 3, at **0.36 dB less** photometric cost. 3.5σ on the prior-to-prior difference.
- *sponza* 1.6σ , *lone-monk* 0.0σ : **two nulls, no regression**. One scene carries this result.

CONCLUSION The prior was the limiting factor on the one scene that was barely helped. This was predicted to be a null; it was not.

Experiment 7: the DA3 family against ground truth

EXPERIMENT Each prior against reference depth, given increasing freedom to fit: none, one scale per frame, then an affine per frame and per patch. Alignment is fitted *in the prior's own quantity* and against ground truth, so every number is an upper bound on what a sparse-point alignment could reach.

Prior accuracy against ground truth by alignment freedom (lower is better)



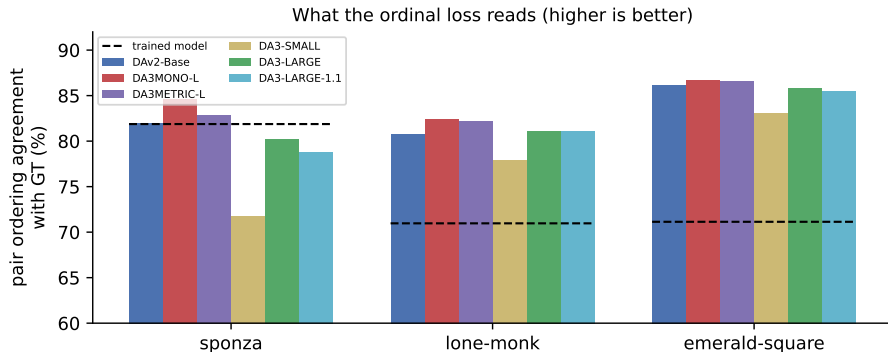
prior	sponza				lone-monk				emerald-square			
	raw	scale	affine	16px	raw	scale	affine	16px	raw	scale	affine	16px
DAv2-Base (mono)	0.9485	0.4454	0.0646	0.0107	0.9748	0.2488	0.0433	0.0121	1.0113	2.1182	0.1228	0.0357
DA3MONO-L (mono)	0.8364	0.1182	0.0511	0.0103	0.8694	0.0522	0.0365	0.0116	0.9465	0.1293	0.0854	0.0311
DA3METRIC-L (mono, metric)	0.5489	0.0833	0.0649	0.0104	0.3913	0.0457	0.0395	0.0115	0.4105	0.0943	0.0811	0.0316
DA3-SMALL (any-view)	0.8516	0.2536	0.2160	0.0157	0.8888	0.0905	0.0911	0.0165	0.9534	0.1938	0.1743	0.0462
DA3-LARGE (any-view)	0.8637	0.1164	0.0967	0.0119	0.8873	0.0424	0.0396	0.0128	0.9157	0.1076	0.0976	0.0377
DA3-LARGE-1.1 (any-view)	0.8657	0.1232	0.1008	0.0122	0.8896	0.0421	0.0414	0.0130	0.9333	0.1120	0.0972	0.0380

Experiment 7: reading the ladder

RESULT

- **Unfitted, all the relative priors are useless** (abs_rel 0.84–1.01) and this says nothing about them: COLMAP's scale is arbitrary, so the row measures the unit mismatch. `DA3METRIC-L` reaching 0.39–0.55 means only that COLMAP's units land within about $2\times$ of metres here.
- **One scale is enough for DA3, and not for DAv2.** Denied an offset, DAv2 gives 0.25–2.12 — on `emerald-square` *worse than no fit at all* — while the DA3 monocular models give 0.05–0.13. A disparity prior needs its shift term; a depth prior does not. This is the rung a sparse-point alignment would actually live on.
- **Any-view checkpoints are worse monocular priors.** `DA3-LARGE` loses to `DA3MONO-L` on every scene and rung, and on `sponza` loses to DAv2 as well (0.0967 vs 0.0646, one affine per frame). Handed one image they cannot do the thing they exist for. `DA3-LARGE-1.1` is indistinguishable from `DA3-LARGE`.
- `DA3-SMALL` **is worse than everything, everywhere**, by $2\text{--}4\times$ at the global rung. Size matters within the family.
- By 16×16 patches all priors converge to 0.010–0.046: local structure is common to all of them, and the differences above are about global consistency.

Experiment 7: what the ordinal loss actually reads

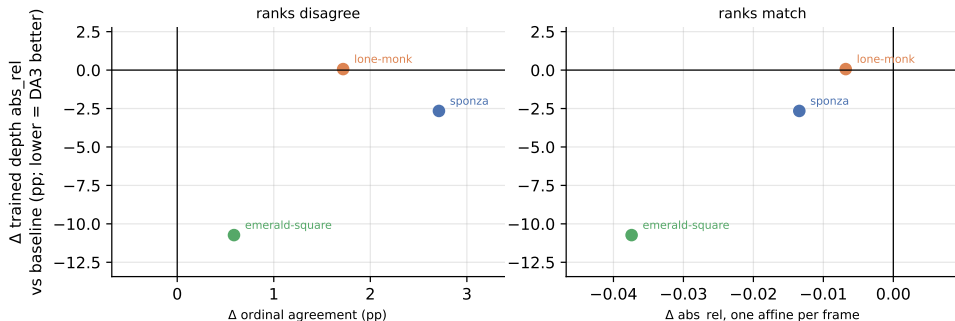


- DA3MONO-L orders best on all three scenes; DA3METRIC-L is within 0.2 pp of it on two.
- DA3SMALL **falls below the trained model on** *sponza* (71.7% vs 81.9%). Supervising with it would teach the model orderings it already gets right — a prior can be net harmful, and this is the check that says so before a sweep is spent.

CONCLUSION DA3MONO-LARGE is the best available monocular prior on every offline measure taken, and it is the largest *monocular* checkpoint published — larger DA3 models exist only in the any-view family, which measured worse here.

Experiment 7: which diagnostic would have predicted this?

Does the offline diagnostic rank the scenes the way training did? (DA3MONO – DAv2)



- Ordinal agreement is *what the loss consumes*, and it ranks the scenes wrongly: `emerald-square` gained the least agreement (+0.6 pp) and the most depth (–11 pp), `sponza` the most agreement and little depth.
- Accuracy under **one affine per frame** — a quantity the loss never sees — ranks all three correctly.
- Hypothesis going in was that training gain would track ordinal agreement. Refuted.

CONCLUSION Three scenes and one prior swap: enough to reject “agreement is the number to optimise”, not enough to install its replacement. Treat the diagnostic as a *screen* — it caught DA3-SMALL — and let the ablation decide.

Experiment 8: what killed regression was the mask, not the idea

PREMISE Experiment 3 abandoned regression on one number: the aligned prior at `abs_rel` 0.0677 against the model's 0.0580. Experiment 7 showed the prior was not the one to measure. Two further problems with that screen:

- **One scene.** On `lone-monk` and `emerald-square` a sparse-point-aligned DA3MONO-LARGE is **2.2–2.4× better** than the 7k model (0.042 vs 0.091, 0.079 vs 0.188). Only `sponza` goes the other way.
- **One mask.** The screen scored both prior and model on `confident & depth>0 & valid`. The metric the ablation reports uses a looser one, on which the same `sponza` baseline is **0.061, not 0.037** — and the prior, at 0.058, is better there too. **The verdict flipped on the mask.**

EXPERIMENT Fit one affine per frame against the COLMAP sparse points — no ground truth — then L1 in ray-distance space, masked on accumulated opacity. 3 seeds, `trisurfel`, DA3MONO-LARGE. Alignment quality is verified *through the training path*, pinned by a test: a wrong downscale factor gives a smooth, plausible, uniformly wrong depth map. **RESULT**

Run	<i>sponza</i>			<i>lone-monk</i>			<i>emerald-square</i>		
	<code>abs_rel</code>	δ_1 fail	PSNR	<code>abs_rel</code>	δ_1 fail	PSNR	<code>abs_rel</code>	δ_1 fail	PSNR
<code>trisurfel</code> (ref)	0.0610	5.5%	36.07	0.0991	18.7%	36.74	0.1282	17.5%	32.76
pd 0.1, DA3MONO-L	0.0523	4.8%	36.13	0.0867	17.7%	36.65	0.1136	13.3%	32.08
pd11 0.01	0.0564	4.5%	36.02	0.0476	3.8%	36.70	0.0713	6.3%	33.13
pd11 0.1	0.0544	3.7%	36.41	0.0396	1.8%	36.51	0.0695	5.9%	32.43
pd11 1	0.0658	5.0%	34.95	0.0425	1.9%	35.03	0.0712	5.8%	30.32

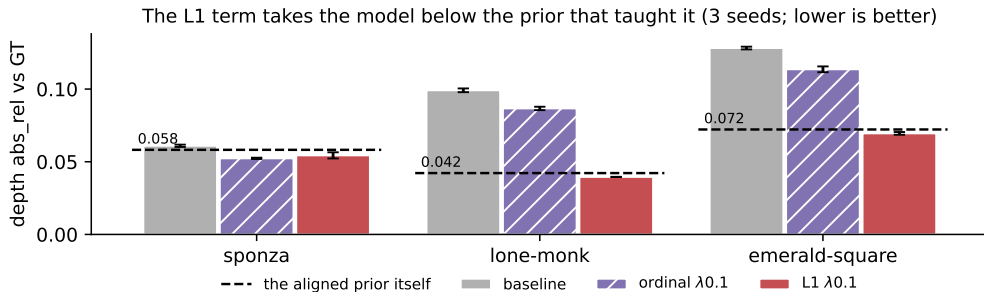
Experiment 8: result

- -60% / -46% / -11% `abs_rel` at $\lambda = 0.1$ on `lone-monk` / `emerald-square` / `sponza`. **$4-5\times$ any other term in this report**, and every scene improves — including the one predicted to regress.
- δ_1 moves further than `abs_rel`: `lone-monk`'s failure rate goes $18.7\% \rightarrow 1.8\%$, `emerald-square`'s $17.5\% \rightarrow 5.9\%$.
- **It reaches** `lone-monk`, the scene with no floaters whose depth is wrong in an opaque, confidently-placed way. Every ray-concentration term in this report is structurally unable to touch it; the ordinal form managed only -12.5% .
- PSNR: -0.23 / -0.33 / $+0.34$ dB. $\lambda = 1$ costs $1.1-2.4$ dB, is worse on depth than $\lambda = 0.1$ on all three scenes, and on `sponza` is worse than the baseline outright — **out**. $\lambda = 0.01$ is the better trade on `emerald-square` alone (-44% for $+0.37$ dB).
- Normals move only $0.8-4.3^\circ$. This is a depth term.

CONCLUSION **Landed**. $\lambda = 0.1$, and it supersedes the ordinal term rather than joining it.

Experiment 8: the student beats the teacher

PREMISE The aligned prior is closer to truth on only 65–77% of pixels (p_{closer}), so a quarter to a third are pulled the *wrong* way: the term should be a floor, not a teacher, and could not beat its own prior.



RESULT Refuted on all three scenes. **CONCLUSION** Per-frame prior error is not shared between frames, so fitting one geometry to all of them averages it out and the wrongly-pulled pixels are outvoted. p_{closer} **is not a ceiling**, and a multi-view training signal cannot be reasoned about one frame at a time.

Experiment 8: do the two forms of the prior compose?

PREMISE The ordinal term reads only the prior's ordering and is invariant to the alignment; the L1 term reads its values and depends on the alignment entirely. If L1's gain were really just better ordering, adding the ordinal term should be free or better.

Run	<i>sponza</i>			<i>lone-monk</i>			<i>emerald-square</i>		
	abs_rel	δ_1 fail	PSNR	abs_rel	δ_1 fail	PSNR	abs_rel	δ_1 fail	PSNR
trisurfel (ref)	0.0610	5.5%	36.07	0.0991	18.7%	36.74	0.1282	17.5%	32.76
pd 0.1, DA3MONO-L	0.0523	4.8%	36.13	0.0867	17.7%	36.65	0.1136	13.3%	32.08
pd11 0.1	0.0544	3.7%	36.41	0.0396	1.8%	36.51	0.0695	5.9%	32.43
pd 0.1 + pd11 0.1	0.0548	3.5%	36.20	0.0394	1.8%	36.47	0.0667	5.4%	32.35

- Together they are indistinguishable from L1 alone: -0.0028 on *emerald-square*, -0.0002 on *lone-monk*, $+0.0004$ on *sponza*. Seed noise is ± 0.002 – 0.004 , so all three are inside it, and PSNR is 0.1–0.2 dB worse.
- So the L1 gain is **not** better ordering: once absolute placement is supervised, ordering adds nothing. The two terms are not complementary, they are alternatives, and the cheap one to keep is the one that measured better.

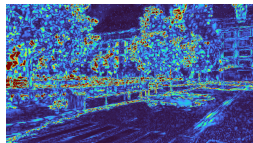
CONCLUSION Ship L1 alone. The ordinal term stays as the fallback for a *disparity* prior, which cannot be aligned in z at all and which the regression path refuses rather than guess.

The photometric cost, localised

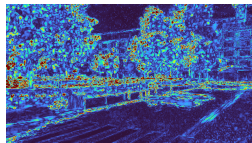
EXPERIMENT PSNR is one number per frame and does not say where a difference sits. Difference every render against the reference image; same colour cap (10% of full range) on every map.



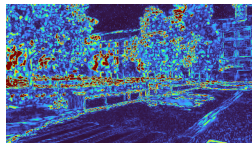
reference RGB



gaussian 32.90 dB



trisurfel 32.92 dB



pd + dn **31.96 dB**

RESULT

- Error in all three is concentrated in **foliage and the hedge line**; the road, buildings and sky are near zero everywhere.
- $pd + dn$ replaces the references' fine speckle with **larger coherent blobs** in the canopy: mean absolute RGB error $4.28 \rightarrow 4.80$ of 255, closer to the reference on only **41%** of pixels.

CONCLUSION The cost is a real regression, not a metric artefact, and it is spatially specific — vegetation, where a consistency term assuming a locally planar surface should be expected to pay most.

The photometric cost, zoomed

QUALITATIVE Canopy crop of the same frame, against the better reference run here by PSNR. `gaussian` carries the same artifacts and is omitted for space.



reference



trisurfel 32.92 dB



pd + dn 31.97 dB

RESULT Two changes at once:

- The references' **spiky starburst artifacts** (left of frame) are **gone** — they are not in the reference image either.
- Genuine foliage structure is **smoothed away** with them, and against the reference that costs more than the artifacts did.

CONCLUSION Removing visible artifacts and losing PSNR are not contradictory, but here the loss is not paid for: the render moves away from the reference on the majority of pixels.

Summary: depth

RESULT `abs_rel`, and the change against each scene's **best reference run**.

Run	sponza		lone-monk		emerald-square	
gaussian (ref)	0.0571		0.0982		0.1368	
trisurfel (ref)	0.0600		0.0987		0.1327	
pd 0.1	0.0506	−11.4%	0.0876	−10.8%	0.1228	−7.4%
dn 0.05	0.0474	−17.0%	0.0976	−0.5%	0.1309	−1.4%
dvrel 0.01	0.0526	−7.9%	0.0993	+1.2%	0.1325	−0.2%
pd 0.1 + dn 0.05	0.0472	−17.3%	0.0912	−7.1%	0.1262	−4.9%
pd 0.1 + dvrel 0.01	0.0492	−13.9%	0.0880	−10.4%	0.1275	−3.9%

- `pd 0.1` is the only *single* term improving depth on all three scenes by more than seed noise (± 0.002 – 0.004). `dn 0.05`'s `lone-monk` and `emerald-square` gains (0.0005, 0.0018) are inside it; `dvrel` loses on `lone-monk`. Both pairs clear it on all three, at single seed.
- Best per scene: `pd + dn` on `sponza`, `pd` alone on `lone-monk` and `emerald-square`.
- These all use the `DepthAnythingV2-Base` prior. Swapping it for `DA3MONO-LARGE` (Experiment 7) takes `emerald-square` from −5.3% to −16.0% at less photometric cost.
- **None of these is the best result in the report.** The sparse-aligned *regression* term of Experiment 8 — the loss this table's family was chosen over — reaches −60% / −46% / −11%, 4–5× everything above, and is the only entry that reaches `lone-monk`. The best depth numbers here came from changing the *prior* and then the *form of the loss*; the weight sweeps in Experiments 4–6 moved far less than either.

Summary: normals

RESULT Mean angle, and the change against each scene's **best reference run** (which is `trisurfel` on all three).

Run	sponza		lone-monk		emerald-square	
gaussian (ref)	52.4		54.3		65.8	
trisurfel (ref)	44.7		35.0		49.4	
pd 0.1	52.5	+17.3%	53.1	+51.7%	66.2	+34.1%
dn 0.05	25.4	-43.1%	37.4	+6.7%	41.3	-16.3%
dvrel 0.01	52.4	+17.3%	53.2	+52.1%	65.7	+33.2%
pd 0.1 + dn 0.05	24.0	-46.3%	28.3	-19.2%	30.5	-38.2%
pd 0.1 + dvrel 0.01	52.6	+17.7%	53.1	+51.6%	67.3	+36.4%

- `pd` and `dvrel` leave normals unchanged; both act on depth only.
- `pd + dn` is best on all three scenes, and the only configuration with positive `n_gain` everywhere.

Status and settings

Term	Status	Verdict
Depth-normal consistency	landed, off	Recovers normals; lone-monk still loses to trisurfel
Depth variance, absolute	refuted	Confidence without accuracy; wrong tight $\times 170$
Depth variance, relative	landed, off	Depth on sponza only; null elsewhere, no cost
Scale-z (PGSR port)	landed, small	Only after fixing the normal-axis assumption
Pseudo-depth ordinal	landed, off	Superseded by regression below; keep for disparity priors
+ confidence gate	refuted	Keeps the prior's worst structure
DA3 prior swap	landed, off	$3\times$ the depth gain on emerald-square, 2 nulls
Pseudo-depth regression	landed, off	Best depth here: $-60/ -46/ -11\%$. Abandoned twice, wrongly
Multi-view consistency, mesh	not started	

- Depth, one term: `pd1 0.1` with a `DA3MONO-LARGE` prior. $-60 / -46 / -11\%$ against the best reference, -0.33 to $+0.34$ dB. Needs `align_to_sparse_points`, which is enabled by the loss.
- Depth and normals: `pd 0.1 + dn 0.05`. Best normals everywhere, -0.22 to -0.95 dB. Whether `pd1` substitutes for `pd` in that pair is **untested** — the regression term barely moves normals alone, so the combination is the open question.
- Best prior: `DA3MONO-LARGE`. Not the default only because it needs the `depth_anything_3` package on `PYTHONPATH`, where `transformers` installs with the venv. Both are Apache-2.0.
- All remain **off by default**: the photometric cost is real and PSNR is what the codebase is currently evaluated on.

Limitations

- **7k iterations only.** No term confirmed at 30k. Iteration-gated defaults (`*_from_iter`) were lowered for the sweep and do not transfer unchanged.
- **Three scenes**, chosen because they fail differently, not because they are representative.
- **Combination cells are single-seed** against 3-seed references. The `lone-monk` `n_gain` sign flip and the `sponza` depth gains are far outside seed noise; `emerald-square` differences of ~ 1 point are not.
- **Why `emerald-square` pays most is unexplained.** That the cost is concentrated in foliage is measured; the link to its depth range is a hypothesis.
- **No mesh.** Nothing here is evaluated by surface extraction or Chamfer distance.
- **The DA3 result rests on one scene.** `emerald-square` is 3.5σ ; `sponza` and `lone-monk` are nulls. Why `emerald-square` is the scene where the prior binds is unexplained.
- **The unfitted rung measures units, not priors.** COLMAP scale is arbitrary, so it cannot rank relative priors and is reported only to show what alignment supplies.
- **Any-view checkpoints were run on single images**, which is not what they are for. Their pose-conditioned path is untested here.

Limitations (cont.)

- **Experiment 7's prior alignment is fitted against ground truth**, so every prior abs_rel there is an upper bound. Experiment 8's is not: it fits the COLMAP sparse points, which costs +11 to +14% on two scenes and is 4–7% *better* on the third, so “upper bound” was itself wrong — a least-squares fit over all pixels does not bound a metric that weights near pixels more.
- **Experiment 8 needs a depth prior and a COLMAP reconstruction.** The alignment is an affine in z ; a disparity prior is affine to *inverse* z , and the dataset raises rather than fit the wrong thing. Datasets other than `colmap` have no sparse points and cannot use the term at all.
- **Sky is unhandled.** A per-pixel depth term wants a sky mask; `DA3METRIC` emits one and `DA3MONO` does not. None of these scenes is sky-dominated, so this is untested rather than solved.
- **Experiment 8 is not combined with anything.** It is measured alone and against the ordinal term; its interaction with dn — the term that supplies the normals — is the largest untested combination in this report.

Method notes that changed a conclusion

1. **Include a control that uses no information.** `gaussian`'s normal buffer looked reasonable and lost to pointing every normal at the camera; `trisurfel`'s looks good, scores 19° better, and still only ties the control.
2. **Report per scene.** `lone-monk` has no floaters and the worst depth; a whole family of terms cannot help it, and a scene mean reported a win.
3. **Compare against the strongest reference, not the one you modified.** Every treatment here is a change to `gaussian`, so `gaussian` was the natural baseline — but `trisurfel` is better on normals everywhere and on `emerald-square` depth. Re-scoring against the best of the two removed a claim that `dn 0.05` beat all references on normals, and cut `pd 0.1`'s `emerald-square` depth gain from 10.3% to 7.4%.
4. **Measure the reference quantity before building on it.** The prior diagnostic cancelled a planned loss; the PGSR port regressed normals 30° because “min scale” is not this renderer's normal axis.
5. **An offline metric can motivate the wrong mechanism.** The gate's 84%→97% was real and irrelevant, and ordinal agreement — the quantity the shipped loss reads — ranked the DA3 swap's three scenes wrongly.

Method notes that changed a conclusion (cont.)

6. **Measure combinations at each term's own optimum**, and against each single term, not only against a baseline.
7. **Localise a metric before interpreting it**. The RGB difference maps turned an assumed metric artefact into a measured regression.
8. **Report how much the alignment is doing**. A prior scored only after an affine fit hides which of the two is being measured. DAv2 and DA3MONO are 0.065 vs 0.051 under one affine per frame, and 0.45 vs 0.12 under one scale — a $4\times$ gap that the affine absorbs, and the rung a sparse-point alignment would really run at.
9. **Check the licence of the artefact you measured**, not the one you downloaded first. DA3MONO-LARGE was recorded as CC BY-NC 4.0 and given as the reason it could not ship; it is Apache-2.0, like DAv2. Only DA3's *any-view* checkpoints are non-commercial. The constraint was carried across from a sibling model, and it suppressed the better prior for a reason that did not exist.

Method notes that changed a conclusion (cont.)

10. **A screen that kills a term must use the metric's own pixel mask.** Regression was cancelled on “prior 0.058 vs model 0.037”. Both numbers were right; the model's was on a mask the reported metric does not use, and on that metric the baseline is 0.061 and the prior wins. **The single largest result in this report was suppressed for two experiments by a mask mismatch.** Quote screen and target over the same population, or do not screen.
11. **Do not reason about a multi-view signal one frame at a time.** $p_{\text{closer}} = 0.65$ says a third of pixels are pulled the wrong way, and was read as a ceiling. Averaged over views it is not: the student beat the teacher on every scene.
12. **Record the effective config, not the requested one.** A 200-iteration smoke run left a row in `results.jsonl` still claiming 7000. It merged into a seed average and turned ± 0.0016 into ± 0.2311 — a convincing fake divergence. Now derived from what `train.py` actually receives, and non-7k rows are dropped at plot time.